SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – EXPERIMENTATION

Course Code:	ITA0301	**Course Title:**	Mobile computing
CO Assessed:	CO2	**Assessment:**	Assessment Tool 2 – EXPERIMENTATION
Weightage:	30 % of CIA	**Total Marks:**	20 marks
CO2 Statement:	*Compare mobile communication technologies, including multiple access techniques and network evolution, based on performance and applications.*		
Student Name:	S. Dharmika	**Reg. No.:**	192421094
Section / Batch:		**Date:**	31/7/2026

Code of Conduct

I S.Dharmika/192421094 (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S. Dharmika

Instructions

- This test contains ONE question.
- Duration: 30 minutes.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _______________	Signature: _______________	Signature: _______________
Date: ___________________	Date: ___________________	Date: ___________________

Annexure A - QUESTIONS

1. Figma-Based Mobile IP Communication Workflow Analysis (BL4 – Analyze)

Design an interactive Figma prototype illustrating the complete workflow of Mobile IP communication when a Mobile Node moves from its Home Network to a Foreign Network. Your design should include the Home Agent (HA), Foreign Agent (FA), Correspondent Node (CN), Care-of Address (CoA), Registration Process, and Packet Tunneling.

Analyze the communication process by identifying:

- The purpose of each network component.
- Points where packet delay or packet loss may occur.
- The effect of node mobility on communication performance.
- Possible improvements to reduce latency.

Deliverables:

- Figma (.fig) design
- GitHub repository containing exported images/PDF and documentation (README).

2. Comparative Network Architecture Design Using Figma (BL4 – Analyze)

Using Figma or draw.io, create a comparative architectural diagram showing both a Cellular Wireless Network and a Mobile Ad Hoc Network (MANET) operating in the same smart city environment.

Analyze the architectures by comparing:

- Infrastructure dependency.
- Routing mechanisms.
- Scalability.
- Fault tolerance.

- Mobility support.
- Suitable application scenarios.

Highlight the advantages and limitations of each architecture through annotations in your design.

Deliverables:

- Editable design file.
- GitHub repository with screenshots, comparison report, and design assets.

3. Routing Protocol Visualization and Analysis (BL4 – Analyze)

Develop an interactive visualization using Figma, Mermaid.js, D3.js, or any visualization framework illustrating the operation of one Proactive Routing Protocol and one Reactive Routing Protocol in a mobile wireless network.

Analyze:

- Route discovery process.
- Route maintenance mechanism.
- Routing overhead.
- Delay in establishing communication.
- Performance under high node mobility.

Include a comparative summary explaining which protocol performs better under different mobility conditions.

Deliverables:

- Source code/design files.
- GitHub repository with implementation, screenshots, and analysis report.

4. Mobile IP Application Design for Smart Healthcare (BL4 – Analyze)

Design a Smart Healthcare Mobile IP system using Figma that enables doctors, ambulances, hospitals, and patients to remain connected while moving across different wireless networks.

Analyze:

- How Mobile IP maintains uninterrupted communication.
- Challenges during handoff between networks.
- Security vulnerabilities during mobility.
- Possible design improvements to increase reliability.

Provide navigation between screens demonstrating different mobility scenarios.

Deliverables:

- Interactive prototype.

- GitHub repository containing design files, exported prototype, and documentation.

5. Wireless Network Decision Dashboard Using GitHub Project (BL4 – Analyze)

Create a web-based dashboard (using HTML/CSS/JavaScript, React, or another programming framework) or an interactive Figma dashboard that helps network engineers determine whether a given scenario should use Mobile IP, Cellular Networks, or Ad Hoc Networks.

The dashboard should analyze multiple parameters such as:

- Node mobility.
- Infrastructure availability.
- Network size.
- Routing complexity.
- Energy consumption.
- Reliability requirements.

The system should justify its recommendation based on the analyzed parameters.

Deliverables:

- Source code or Figma project.
- GitHub repository with complete code/design, README, screenshots, and analysis report.

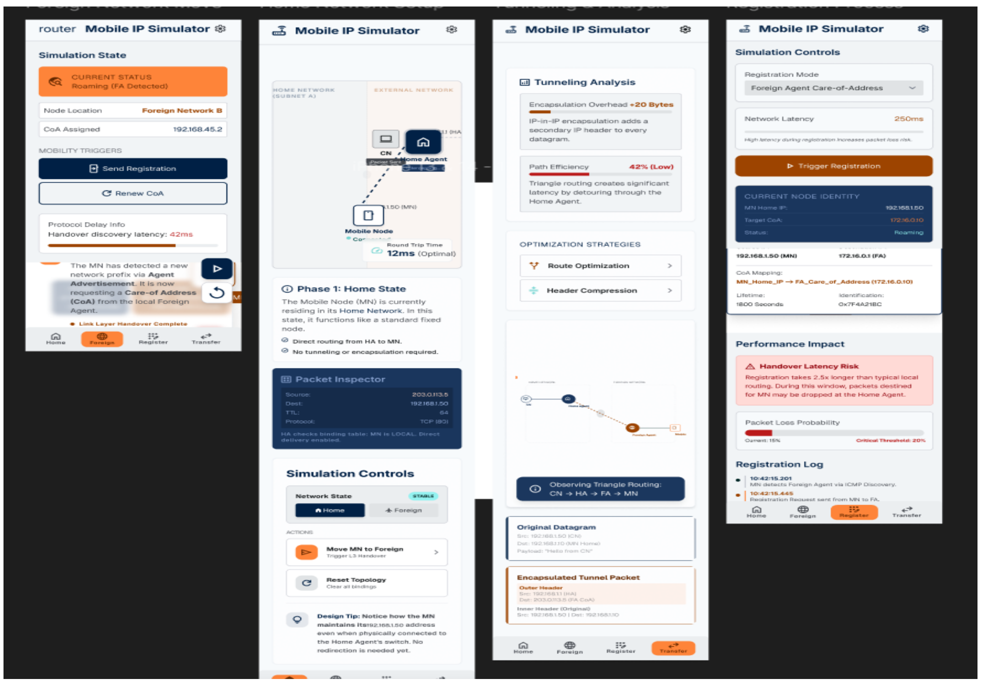
Criterion	Marks
Design Quality (Figma/Programming Tool)	20
Technical Analysis (BL4)	30
Correctness of Mobile Networking Concepts	20
GitHub Repository Organization & Documentation	15
Creativity and Visualization	15
Total	**100 Marks**

Answers

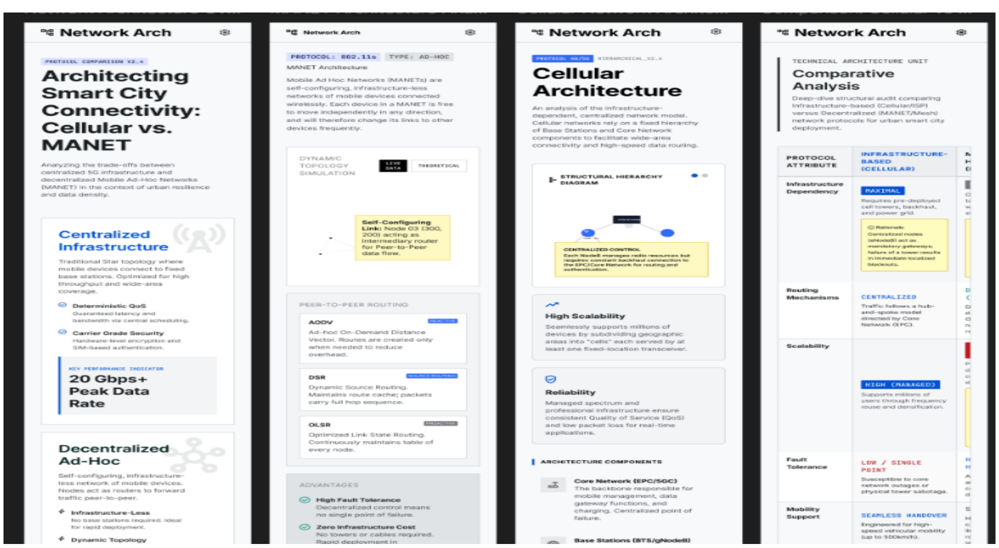
FIGMA LINK:

https://www.figma.com/design/DSk5Ts5aopK3CnKJ39ydO8/Untitled?t=3qIsGvlZNZqIgvd9-0

1)



2)



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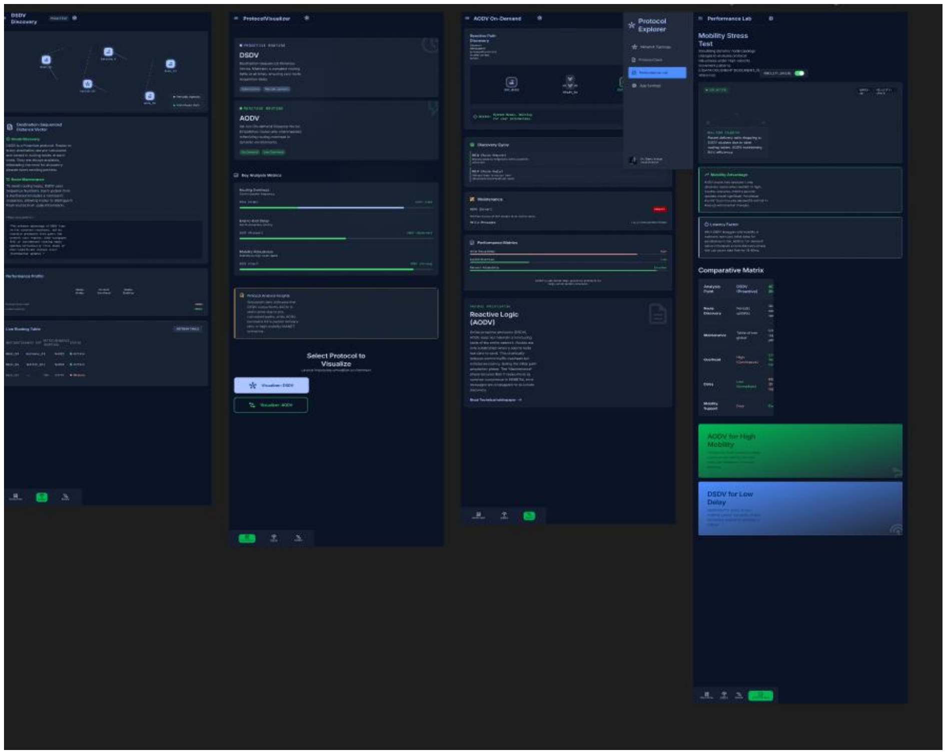

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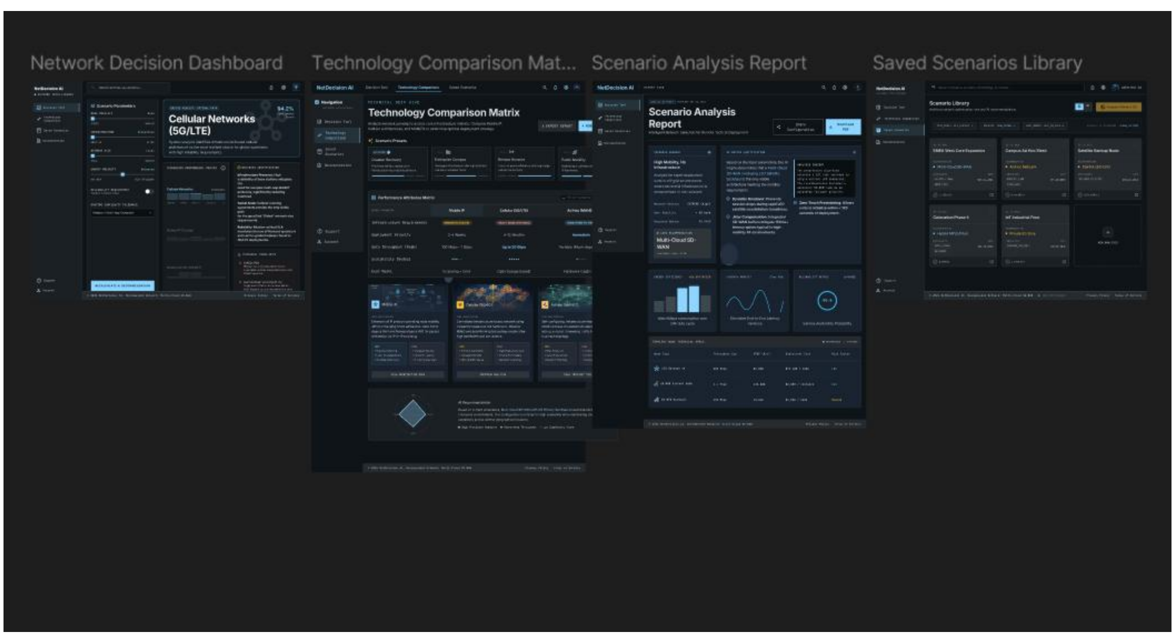
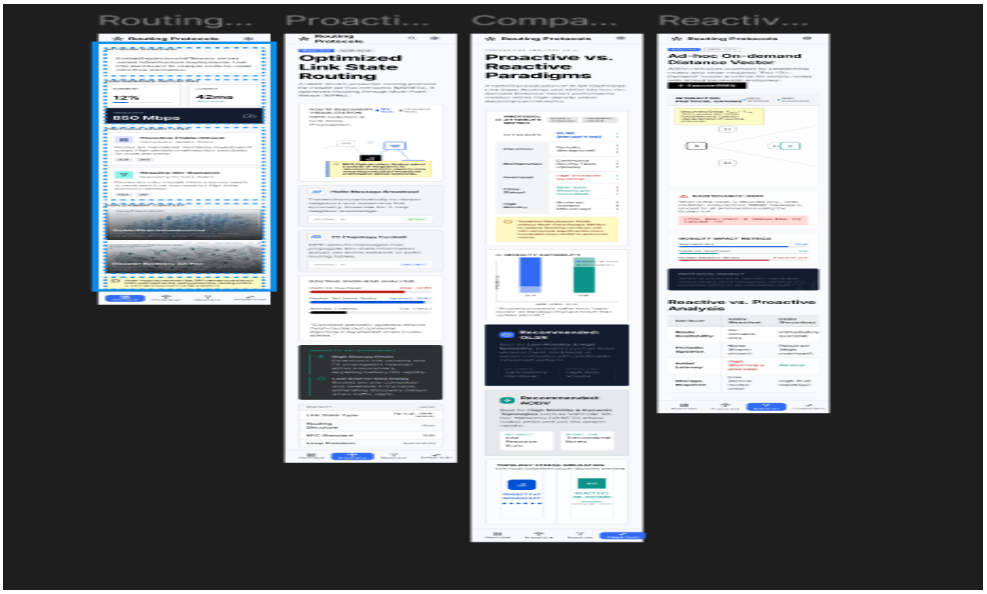

Network Decision Dashboard
Technology Comparison Mat...
Scenario Analysis Report
Saved Scenarios Library

5)


Routing...
Proacti...
Compa...
Reactiv...
Routing Protocols
Optimized Link State Routing
Proactive vs. Reactive Paradigms
Ad-hoc On-demand Distance Vector
850 Mbps
Reactive vs. Proactive Analysis